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1 Summary Tables for Aircraft

1.1 Mission Table

Mission Inputs	EGCC - NTAA	KLAX - KMCO
Range	8,200nm	1,921 nm
Pax per week	1,560	312
Pax per flight	200	200
Bags per flight	200	200
Flights per week	8	2
Payload	25,000 kg	25,000 kg

Table 1.1: Mission table

1.2 Aircraft Characteristics

1.2.1 Outputs

Weights	Our Output
MTOW	264,099 kg
OEW	121,120 kg
MLW	148,130 kg
FUEL MAX	115,970 kg

Table 1.2: Weight outputs

Ranges	Our Output
MPR	3,128 nm
MER/TPR	4,600 nm
MFR	5,740 nm

Table 1.3: Range outputs

1.2.2 Design Inputs

Part of Mission	Our Input
Weight of Passenger	90 kg
Weight of Bag	35 kg
Maximum Payload	46,000 kg
Typical Range	4,600 nm
Primary Diversion Range	1,200 nm
Secondary Diversion Range	200 nm
Loiter Time	1,800 s
Number of engines	2

Table 1.4: Design Inputs

2 Drawings

2.1 Engine Characteristics

The aircraft in this report (Le Plane 381) uses two CF6-80C2 engines and each of them produces a thrust of 232.2-275.6 kN. It is a turbofan engine and has a fan diameter of 9.3 m with an airflow of 790 kg/s. Overall, the pressure ratio is 30.4 and bypass ratio is 5.15. The engine designed for Le Plane 381 can be seen in Figure 1.

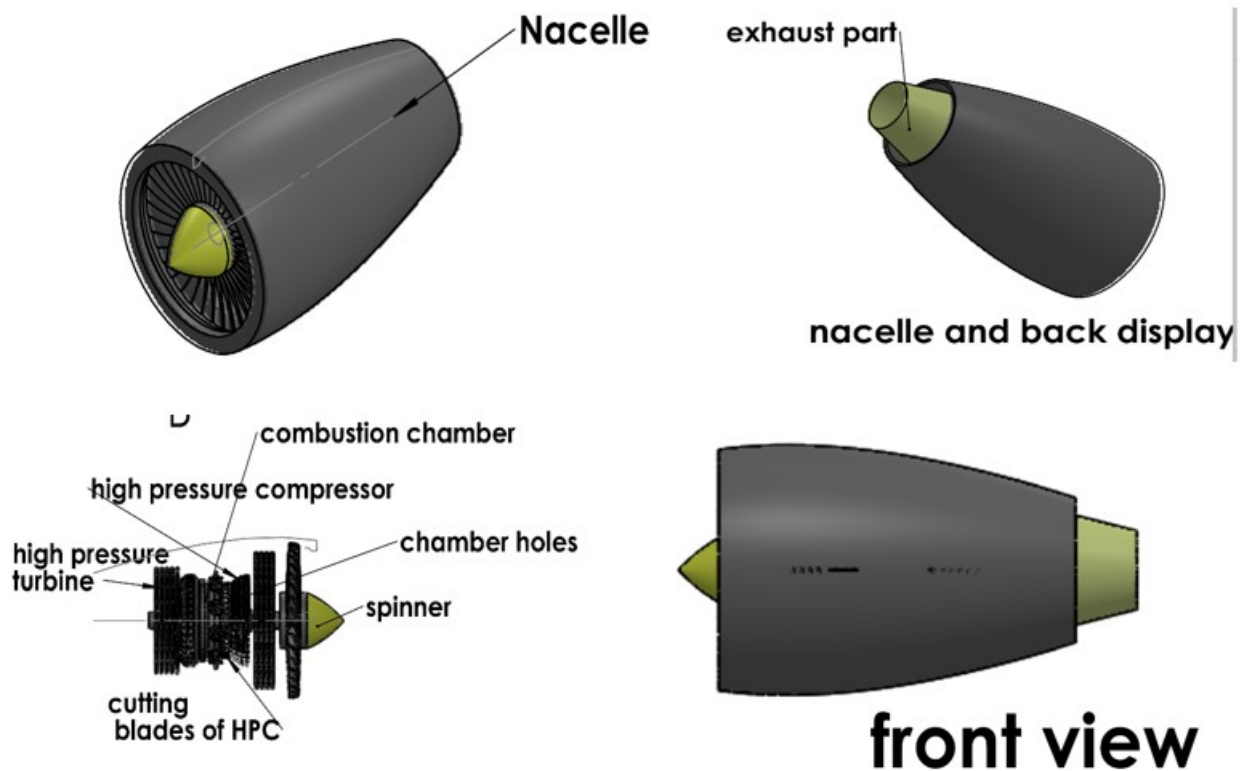


Figure 1: SOLIDWORKS model of engine

2.2 Dimensioned Orthographic and Isometric

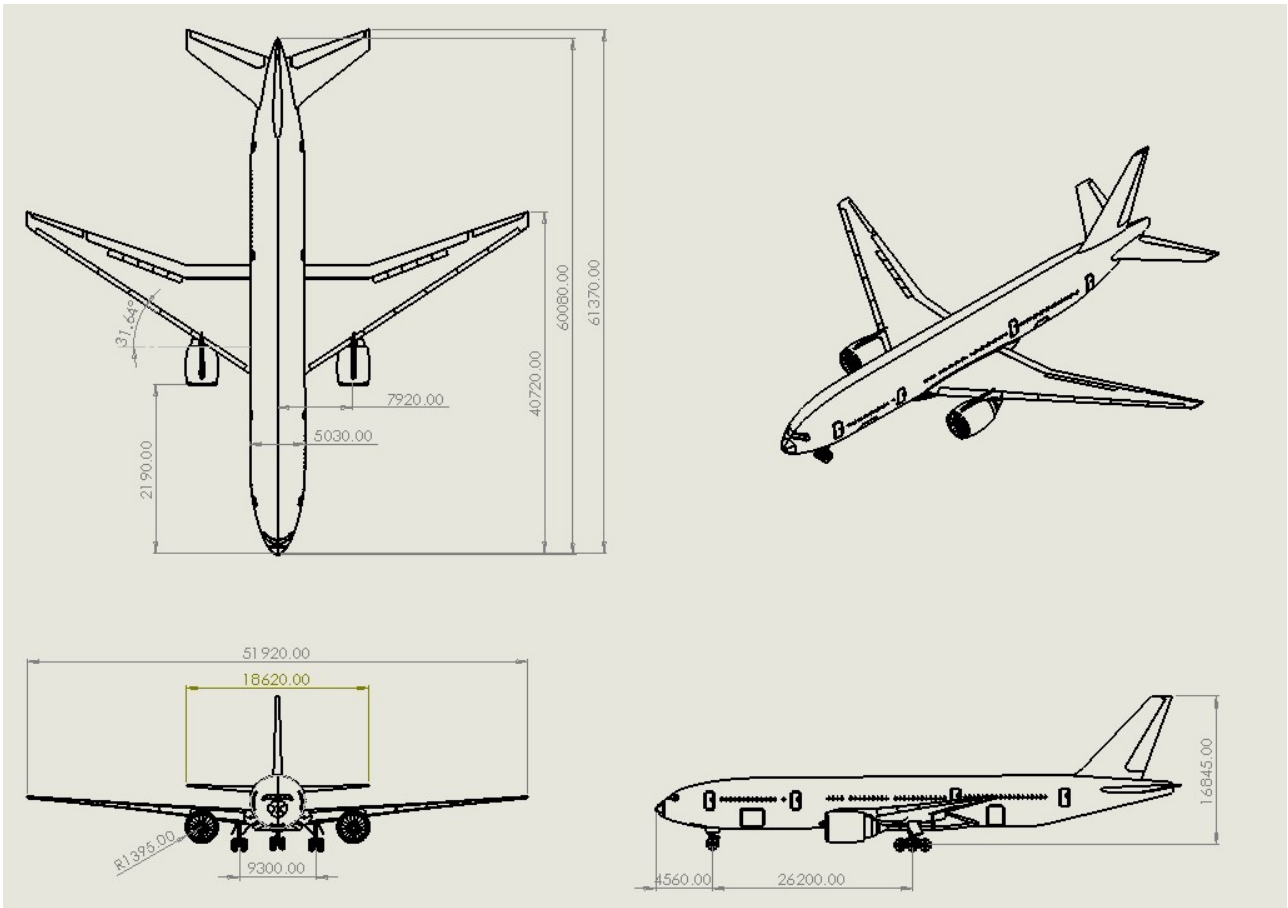


Figure 2: Isometric and Orthographic views of Le Plane 381 [1]

2.3 Seating/Cabin Layouts

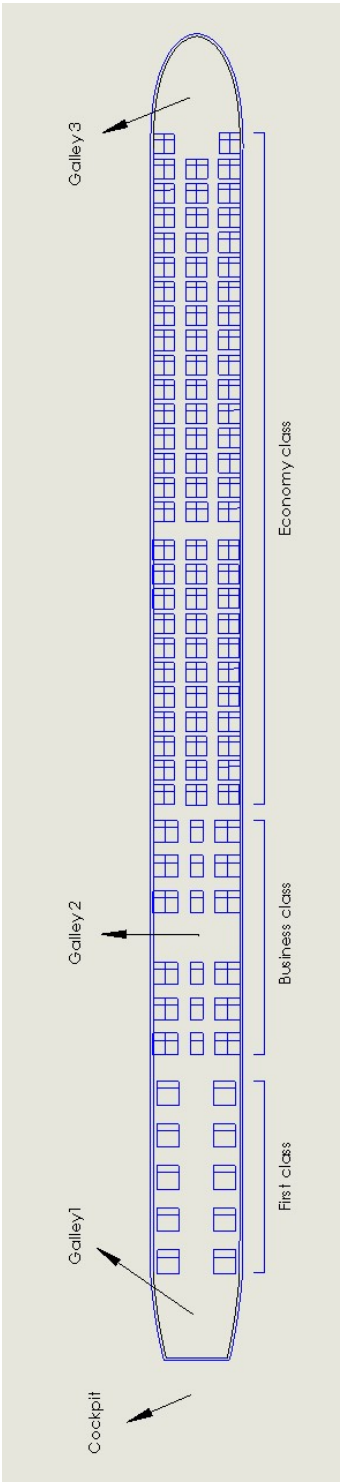


Figure 3: Seating chart

2.4 Cross-sections

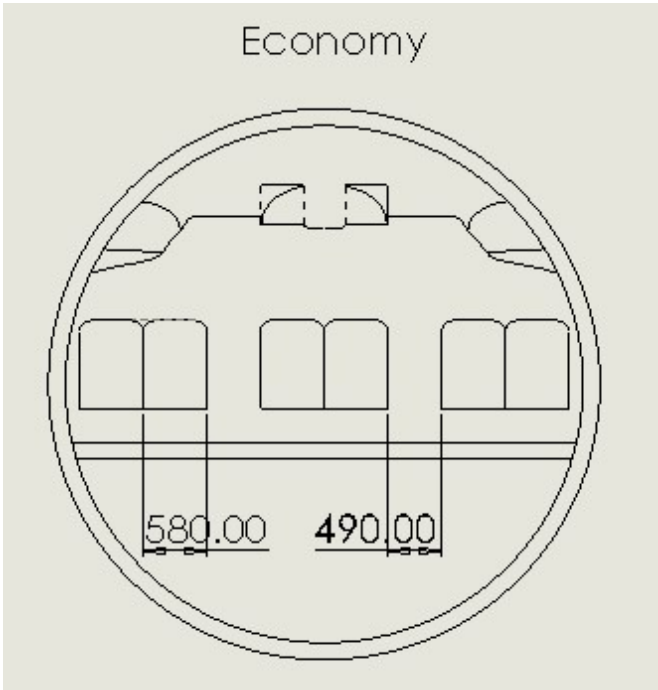


Figure 4: Economy cross section

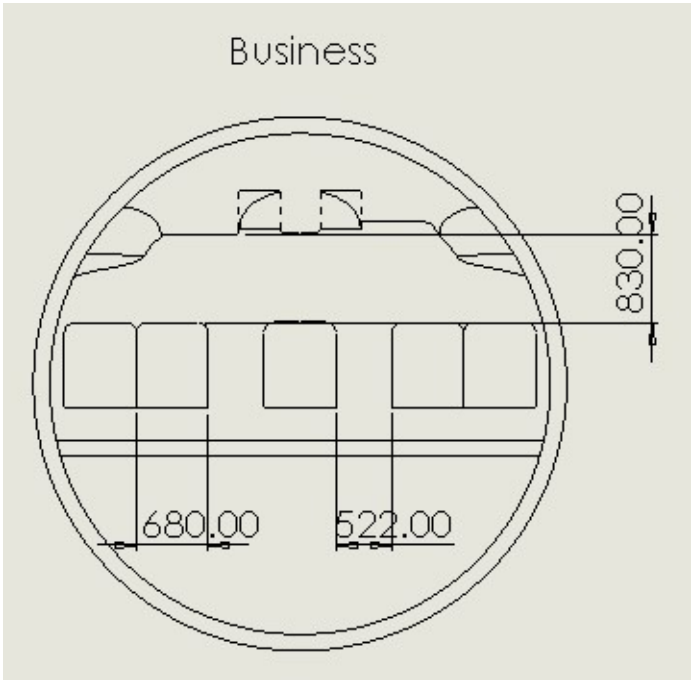


Figure 5: Business class cross section

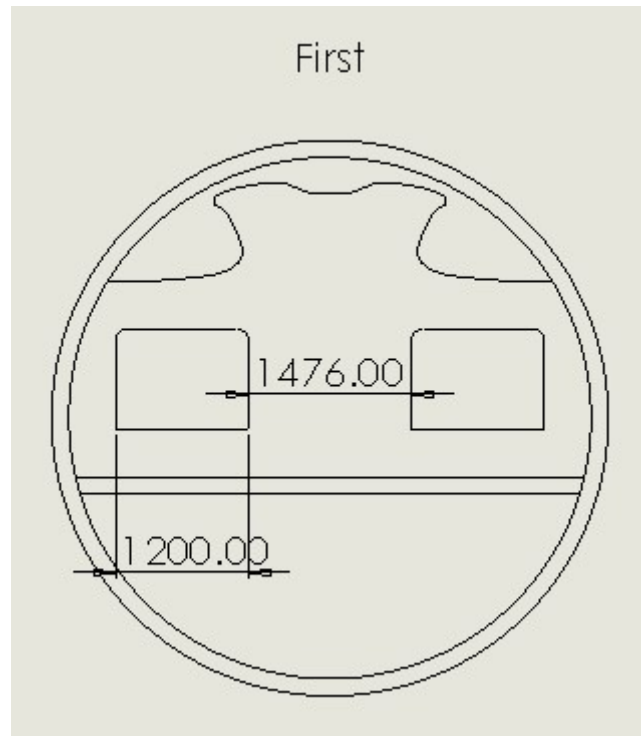


Figure 6: First class cross section

3 Diagrams

3.1 Payload Range

The payload range diagram was obtained using the aircraft design tool programmed in 1st semester. This code marks the maximum payload range (MPR), the typical payload range (TPR) and the maximum ferry range (MFR). The code assumed that the range inputted is the typical payload range, the payload inputted is the typical payload and the rest of the chart is plotted accordingly.

The typical payload range for this chart is taken to be the range from Manchester to Los Angeles, as out of the 2 routes talked about in this report (there is one refuelling stop in Los Angeles for the route from Manchester to French Polynesia), it has the longest range. The maximum payload was assumed to be 46,000 kg. This value was taken from the reference plane for this report, the Boeing 767-400ER.

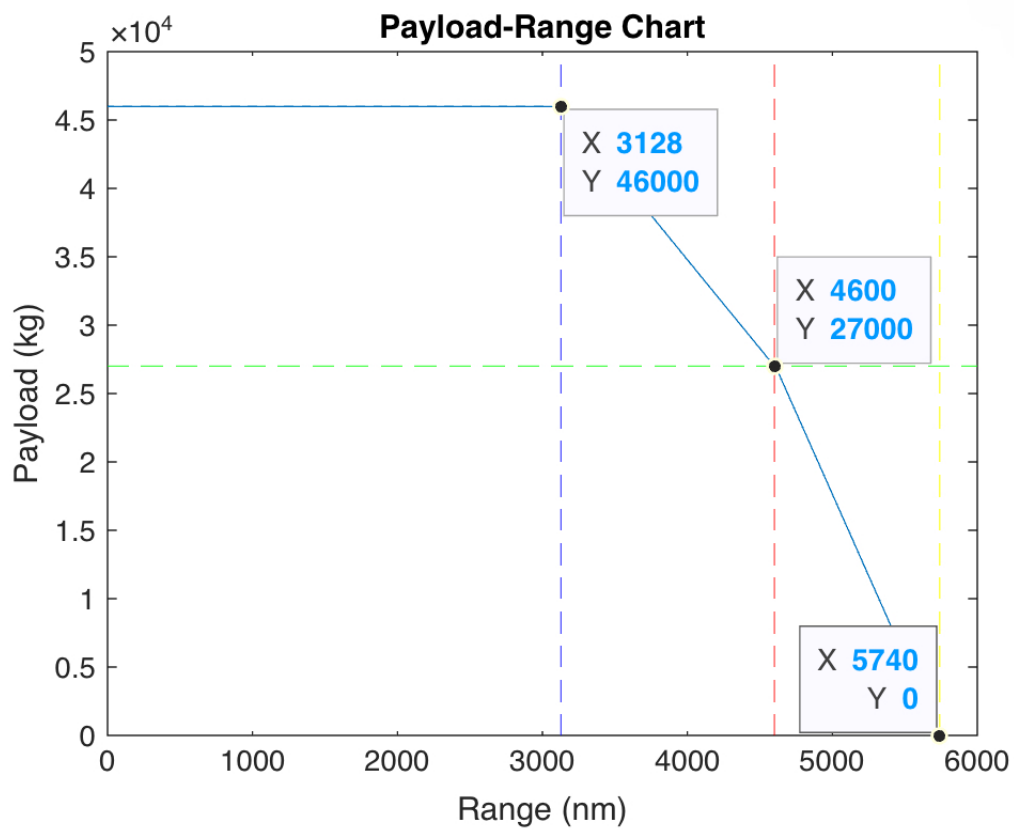


Figure 7: Payload range chart for Le Plane 381

In the payload chart it is clear that even with the maximum possible typical payload range, the plane can very easily fly 3,100 - 5,000 nautical miles with a considerable amount of payload.

3.2 Constraints

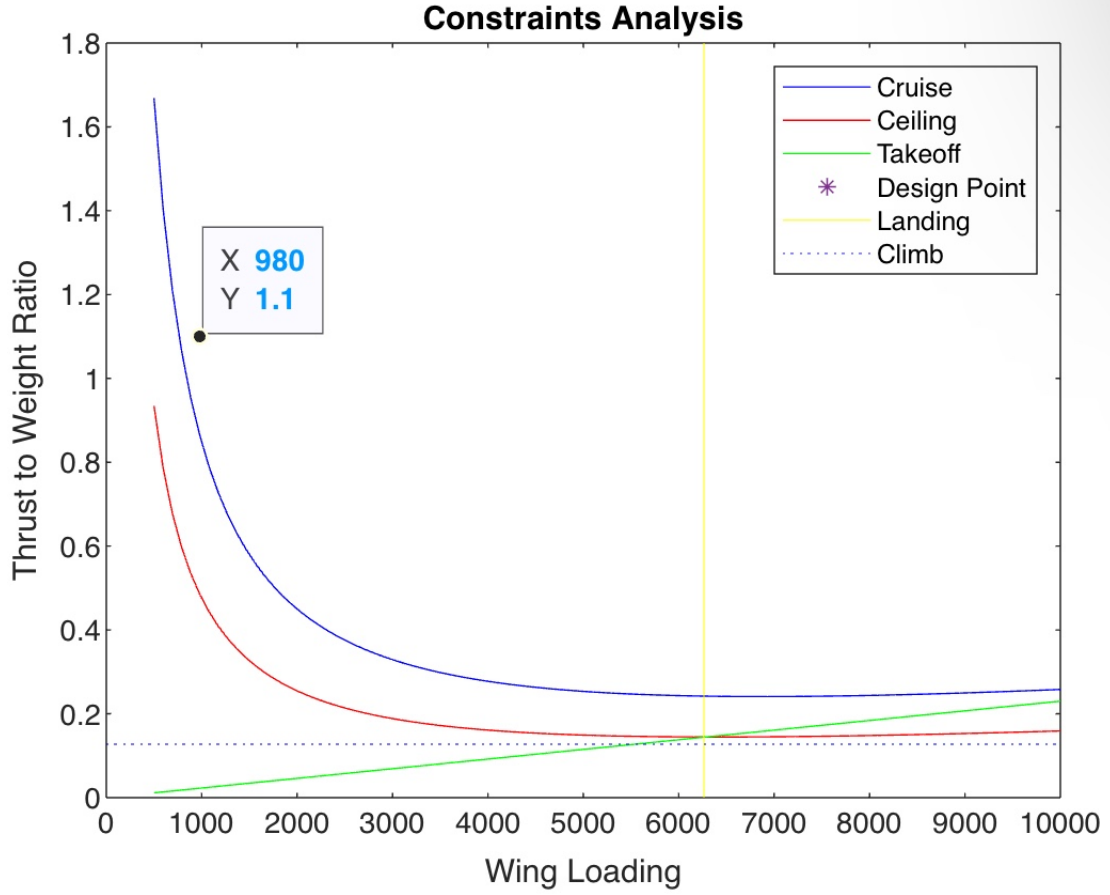


Figure 8: Constraints graph for Le Plane 381

This constraints analysis plot was obtained from the design tool. It plots cruise, climb, ceiling, landing and takeoff constraints for Le Plane 381. The design point illustrates the wing loading and thrust to weight ratio taken for the aircraft, which we can clearly see from the chart are viable values for Le Plane 381. Every constraints plot was done using a form of the master equation (1).

$$\frac{T}{W}\bigg|_{SL,0} = \frac{\beta}{\alpha} \left\{ \frac{qS}{\beta W_0} \left[\kappa \left(\frac{n\beta W_0}{q S} \right)^2 + C_{D,0} + C_{D,R} \right] + \frac{P_S}{V} \right\} \quad (1)$$

In the code, it was assumed that the minimum possible wing loading is 500 Pa and max is 10,000 Pa.

4 Weight and Balance

4.1 CG

The 3D model designed in SOLIDWORKS includes Le Plane 381 in its entirety except the components which would be installed later like the seats, overhead luggage compartments, undercarriage storage etc. A center of gravity value was obtained for this model using the "evaluate mass" tool in SOLIDWORKS at 31.093 m from the tip of the nosecone and 4.553 m above the landing gears bottom.

Center of gravity locations were calculated for the business and first-class seats. The weight of one seat was assumed to be 70 and 100 kg respectively. The same was done for economy seats. However, the economy seats were divided into two sections for ease of calculation and the weight was assumed to be 20 kg.

Overhead luggage compartments' details were found from [2]. They weigh 29 kg each and each one of them is 2.8 m long.

Component	Mass (m)	Cg location from nose (x)
3D CAD model components	114,254.5 kg	31.093 m
First class seats	10 seats x 100 kg = 1,000 kg	7.996 m
Business class seats	30 seats x 70 kg = 2,100 kg	19.055 m
Economy seats section 1	156 seats x 20 kg = 3,120 kg	40.696 m
Economy seats section 2	4 seats x 20 kg = 80 kg	56.61 m
Overhead Luggage compartments Section 1	3.5 units x 29 kg = 101.5 kg	7.996 m
Overhead Luggage compartments Section 2	4 units x 29 kg = 116 kg	19.055 m
Overhead Luggage compartments Section 3	12 units x 29 kg = 348 kg	41.10 m

Table 4.1: Calculations of cg values

The weights for first, business and economy class seats were assumed after comparing many different values online and then agreeing on a reasonable value that falls within the range of these values. Many examples of overhead luggage compartments were found online. The compartment in [2] seemed the most appropriate. The weight and length of one compartment was taken from here. This length was then used to find the number of compartments needed in each class.

The weight of the model was simply calculated by subtracting the sum of weights of everything except the parts included in the model, from the Operating Empty Weight obtained from the design tool.

The center of gravity of everything then can be found using (2).

$$x_{cg} = \frac{\sum(m_i x)}{\sum m_i} \quad (2)$$

Here, x corresponds to the values in column 3 and m to column 2 of Table 4.1. After calculation, this value comes out to be approximately 30.954666 m. This will fall within the limits of the mean aerodynamic chord which starts at about 24 m and ends at around 36 m from the nose, being 12.283 m in length.

In this center of gravity calculation, fuel is not considered. For Le Plane 381, the primary fuel tanks are inside the wings, storing about 72 tons of fuel, while the secondary tanks hold about 40 tons of fuel inside the fuselage. The two secondary tanks are placed across from each other in the fuselage with the center of gravity location right in their middle, ensuring balance. This way the centre of gravity of the aircraft does not change drastically when fuel is added into the secondary tanks.

The center of gravity of the fuel tanks in the wings, if considered, will change the center of gravity location. However, this estimation was performed to find an appropriate location for placement of the wings on the fuselage body. While considering the fuel in the wing could make the location more accurate, the process to do so is time-consuming and hence not viable. This is why this assumption has been made and appropriate considerations are taken to account for the weight of the fuel that would be present in the primary tanks while choosing the location for installation of the wing.

4.2 Static Margin and Neutral Point

According to Anderson [3], in general commercial aviation, typical static margin is 5-10% of the mean aerodynamic chord. Le Plane 381 carries a lot of fuel and will go through big changes in the center of gravity throughout each mission. To accommodate for this, a static margin of 15% was taken. Using (3), the center of gravity location (30.954666 m) and MAC (6.176 m) from our wing sizing analysis (8), static margin and neutral point can be obtained.

$$SM = \frac{(x_n - x_{cg})}{MAC} \quad (3)$$

$$x_n = SM + x_{cg} \quad (4)$$

Then, our neutral point is located at 31.88164 m.

4.3 Wing Sizing

Most of the aerodynamic properties of the wing are fixed. This section includes them as well as parameters which must be calculated, instead of being obtained directly.

Le Plane 381's reference aircraft is a Boeing 767-400ER, therefore its aspect ratio was taken as 9.3 [4]. Taper ratio can be obtained from Figure 9. In order to obtain a corresponding taper ratio value for an aspect ratio of 9.3, induced drag factor must be minimised, then using Figure 9, the value obtained for taper ratio is 0.28.

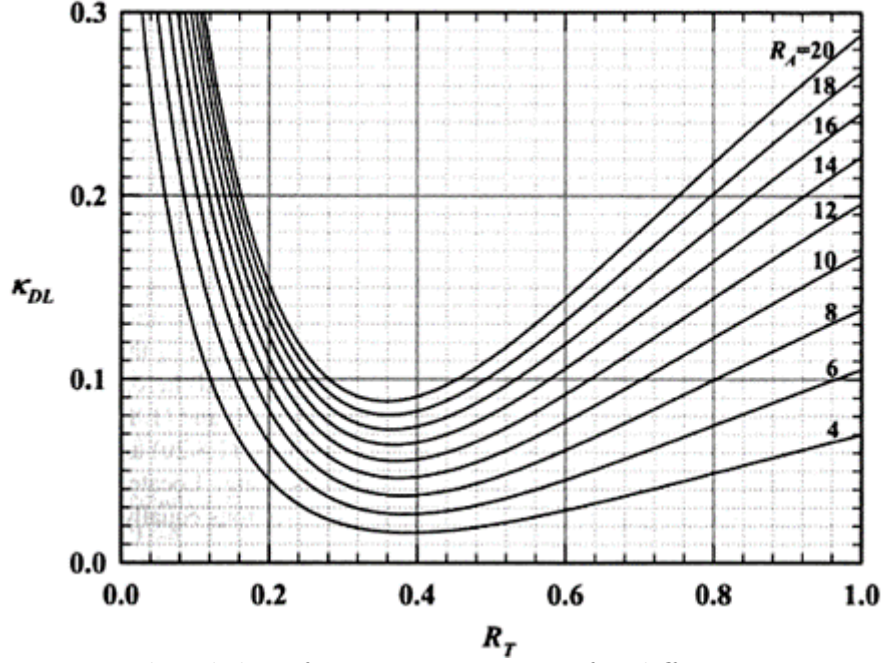


Figure 9: Induced drag factor vs taper ratio for different aspect ratios [5]

Using this value of aspect ratio, wingspan can also be determined using (5).

$$b = \sqrt{AR \times S} \quad (5)$$

- b - wingspan
- S - wing area

This returns a value of $b = 51.995$ m.

The next prominent parameters to be calculated are wing root chord length (c_r) and wing tip chord length (c_t). To find these, average chord length is calculated using (6).

$$c_{avg} = \frac{b}{S} \quad (6)$$

$$c_r = \frac{2 \times c_{avg}}{1 + \lambda} = \frac{2 \times 5.59}{1 + 0.28} = 8.73m \quad (7)$$

$$c_t = \lambda \times c_r = 2.44m \quad (8)$$

- λ - taper ratio

The next step is to find the length of the MAC (Mean Aerodynamic Chord) and its location on the wing (9).

$$MAC = \frac{2}{3} \times \frac{1 + \lambda + \lambda^2}{1 + \lambda} = 6.176m \quad (9)$$

$$y = \frac{b}{6} \times \frac{1 + 2 \times \lambda}{1 + \lambda} = 10.561m \quad (10)$$

The wings of Le Plane 381 have a mid-wing configuration, which provides longitudinal stability. It also distributes the entire aerodynamic load equally, minimising weight and complexity. These wings also have reduced interference drag between the fuselage and the wing, making it more fuel efficient. They provide better ground performance compared to other wing configurations and offer more clearance during ground servicing and take-off/landing operations.

After calculating values for centre of gravity and neutral point, it is decided that it would be ideal to place the wings ahead of the centre of gravity, making them 21 m away from the tip of the nose. This ensures that longitudinal stability is maintained and is the optimal design choice for this aircraft. If any disturbance causes the aircraft's nose to pitch up, then placement of wings here will produce a nose down pitching moment and vice-versa, maintaining the stability. Placing wings here reduces the control efforts of surfaces such as elevators and flaps. The responses to pilot inputs of turn, climb and pitch are outputted earlier when placing wings ahead of centre of gravity. All these points validate placement of wing at 21 metres from the nose of fuselage.

Figure 10 illustrates the wing plot of our designed wing:

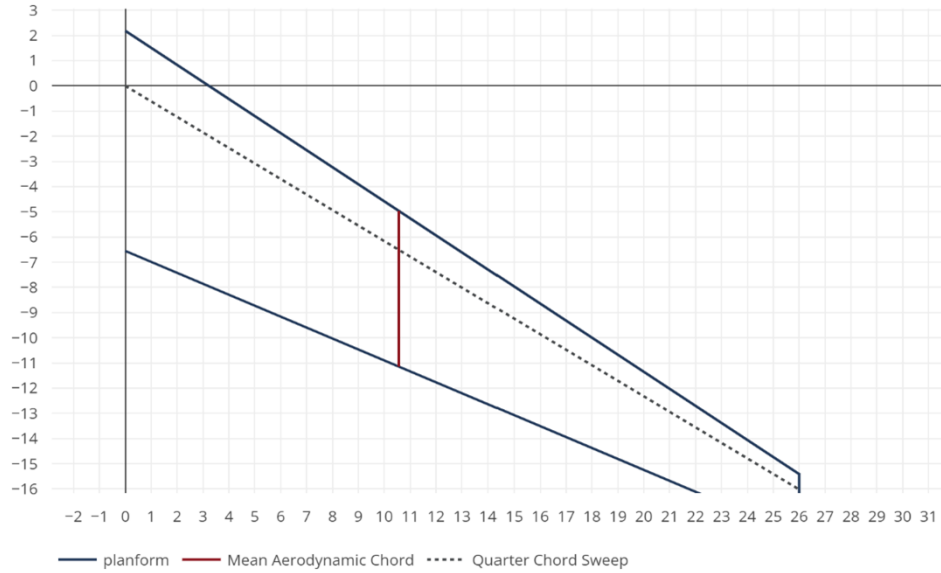


Figure 10: Wing design of Le Plane 381 [6]

Aerofoil considerations

Le Plane 381 cruises at a Mach number of 0.83, therefore the aerofoil is designed so that it can delay the effects of transonic drag as it is about to reach the speed of sound. A thinner aerofoil between a range of 10-15% thickness helps in delaying this onset and improves the aerodynamic efficiency. Also, a supercritical aerofoil has been installed as it offers many advantages when flying long distances, including: delaying the onset of shockwaves in transonic regimes, achieving lower values of drag coefficients and providing higher lift coefficients. The aerofoil incorporated in the design of Le Plane 381 is a NASA SC (2)-0714 which is a supercritical aerofoil, assisting with flap settings and making the aircraft more aerodynamically efficient. This aerofoil is shown in Figure 11.

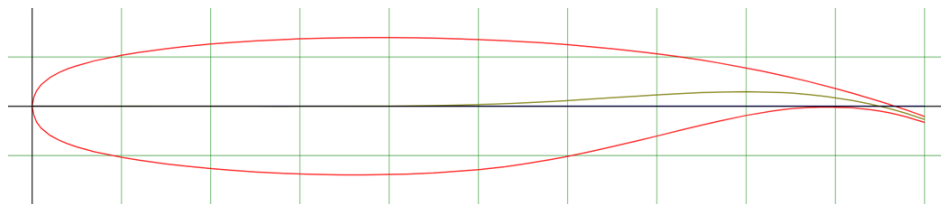


Figure 11: NASA SC (2)-0714 aerofoil [7]

This aerofoil delays the onset of drag divergence by increasing the values of maximum lift coefficient. High lift coefficients increase with increasing angle of attack until a maximum value of lift coefficient is achieved. Beyond this point, stall occurs, as can be seen in Figure 12.

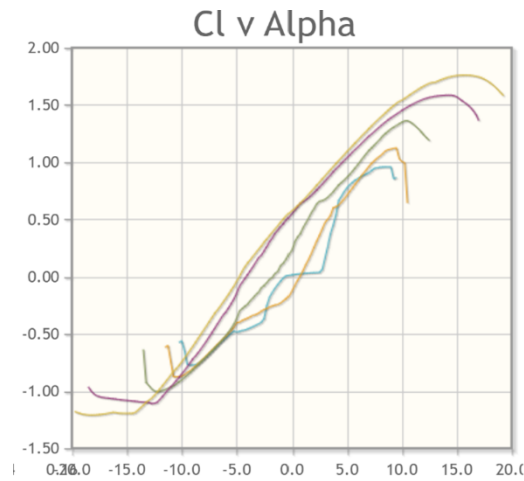


Figure 12: Coefficient of lift vs angle of attack[8]

Flap Settings

The aircraft is also installed with double-slotted flaps which are high-lift devices, usually incorporated on all commercial aircraft, as they offer better aerodynamic efficiencies.

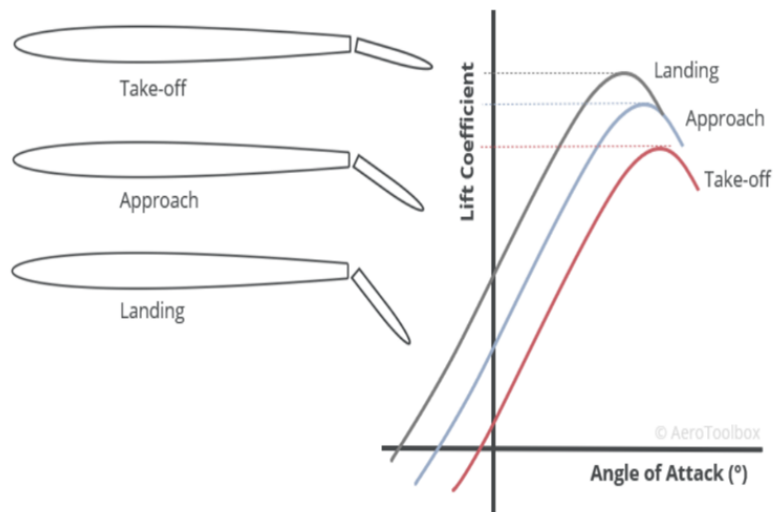


Figure 13: Cl vs AoA for takeoff, approach and landing [9]

Slotted flaps extend downward (see Figure 14) and they are designed in a way so that high pressure air from lower surface moves over the slot and provides high lifting capabilities. Le Plane 381's design uses double-slotted flaps.

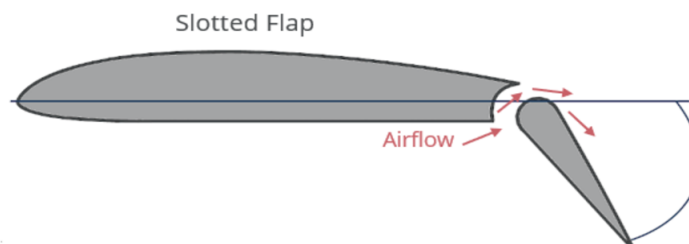


Figure 14: How slotted flaps work [10]

Flaps offer huge advantages during landing as well as takeoff operations. During takeoff,

they reduce stall speed and help the aircraft become airborne quicker. The angle of attack required to maintain a given lift coefficient is also reduced. Climb performance is enhanced by climbing at steeper angles of attack. Landing operations are carried out easily with flap installation, as they make approaching speeds lower by increasing lift coefficient and reducing stall speeds. Instead of single slotted flaps, double-slotted flaps were used, as they create additional opportunities for high pressure air to flow over them, they energise the boundary layer, lift performance is enhanced, wider operating ranges are created and they are safer.

4.4 Tail Sizing

The next important parameter in the design point of the aircraft is tail sizing. This is important, as it provides pitch stability, houses elevators, helping maintaining stability controls and helps when trimming the aircraft. Le Plane 381 includes a T-tail configuration including a vertical stabilizer, horizontal stabilizer, elevators and rudders. The uses of each are as follows:

- Vertical Stabilizer: provides stability in the yaw axis and houses the rudder, which helps to maintain directional stability.
- Horizontal Stabilizer: provides stability in the pitch axis and houses the elevators, helping maintaining stability in that axis.
- Elevators: attached to the trailing edge of the horizontal stabilizer and they control pitch attitude.
- Rudders: attached to the trailing edge of the vertical stabilizer and are used for directional control.

The tail sizing is done based on tail volume ratios, using (11) and (12).

$$V_{HT} = \frac{S_{HT} \times l_{HT}}{S \times \bar{c}} \quad (11)$$

$$V_{VT} = \frac{S_{VT} \times l_{VT}}{S \times b} \quad (12)$$

To find values for the area, the following assumptions are made:

- The horizontal tail volume coefficient is assumed to be 1 (using [11], it falls between 0.5 and 1).

- The vertical tail volume coefficient is assumed to be 0.055 (using [12], it falls between 0.03 and 0.08).
- The moment arm of the horizontal stabilizer (l_{HT}) is calculated by making the assumption that the aerodynamic centre of the horizontal tail is located 58.3 metres away from the tip of the nose. The moment arm would therefore be (13).

$$l_{HT} = x_{cg} - AC_{HT} = 27.35m \quad (13)$$

- The moment arm of the vertical stabilizer (l_{VT}), is also obtained by the assumption that its aerodynamic centre is 55.3 from the tip of the nose. The moment arm would therefore be:

$$l_{VT} = x_{cg} - AC_{VT} = 24.35m \quad (14)$$

All of these values can be substituted into (15) and (16) to obtain the area of horizontal and vertical tails.

$$S_{HT} = \frac{\bar{c}SV_{HT}}{l_{HT}} = 65.64m^2 \quad (15)$$

$$S_{VT} = \frac{bSV_{VT}}{l_{VT}} = 49.65m^2 \quad (16)$$

For the design of the tail of Le Plane 381, a symmetrical aerofoil is incorporated in the design and this helps to provide equal lift characteristics regardless of AOA, they minimise pitching moments and simplify the design of control surfaces. Symmetric aerofoils offer smooth stall behaviour, and this provides easier handling of the control surfaces. The symmetric aerofoil used for Le Plane 381 is a NACA 0012.

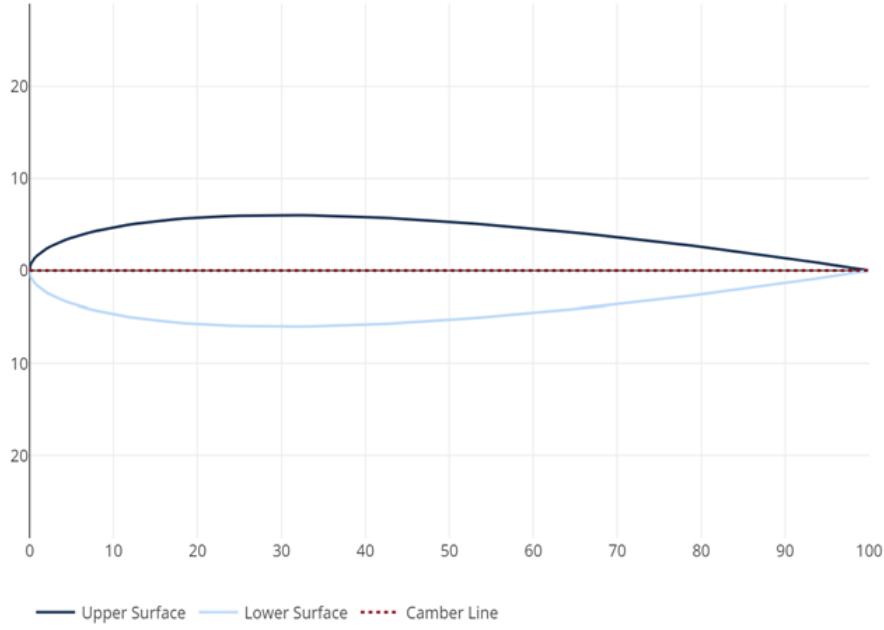


Figure 15: NACA 0012 aerofoil [13]

Obtaining the values of aspect ratio and taper ratio for the tails is a complicated process, therefore, the assumptions shown in table 4.2 were made.

Parameter	Symbol	Value
Aspect ratio of horizontal tail	AR_{HT}	4.46
Aspect ratio of vertical tail	AR_{VT}	1.76
Taper ratio of horizontal tail	λ_{HT}	0.2
Taper ratio of vertical tail	λ_{VT}	0.32

Table 4.2: Assumptions for tail parameters [14]

Based on all these assumptions, the values in Table 4.3 were calculated.

Horizontal tail property	Value	Vertical tail property	Value
b_{HT}	17.11	b_{VT}	9.347
$c_{HT,r}$	6.39	$c_{VT,r}$	8.048
$c_{HT,t}$	1.278	$c_{VT,t}$	2.575
y_{HT}	3.327	y_{VT}	3.87
c_{HT}	4.402	c_{VT}	5.782

Table 4.3: Horizontal tail properties

4.5 More Sophisticated Approaches (Tables 2 & 3)

4.5.1 Detailed Derivation of $C_{D,0}$

To find the total drag experienced by the aircraft, an individual analysis of every component (wing, fuselage, tail surfaces and nacelle) was performed. For civil aircraft, the total drag is categorised into three main types:

1. Parasitic drag: this is the profile drag of a complete aircraft, which is the combination of skin friction and form drag.
2. Induced drag: a pressure drag produced by the pressure imbalance in the drag direction caused by the vortices created at the tips of the wings.
3. Wave drag: drags produced by shockwaves at supersonic speeds.

In this section, only the effects of the profile drag will be considered and its corresponding drag coefficient at zero angle of attack will be calculated. Each component's contribution will be calculated separately using (17) [15].

$$C_{D,0} = C_f \times F \times Q \times \left[\frac{S_{wet}}{S_{ref}} \right] \quad (17)$$

- C_f - skin friction coefficient
- F - component shape factor
- Q - interference factor
- S_{wet} - component wetted area
- S_{ref} - reference area (wing planform area)

Estimating C_f for every component

Initially, Reynold's number must be determined (18), as skin friction coefficient is a function this.

$$Re = \frac{\rho \times v \times l}{\mu} \quad (18)$$

- v - cruise velocity of the aircraft
- l - characteristic length of the component under consideration
- μ - dynamic viscosity

The values to be used in the Reynold's number formula are all taken at 36,000 ft (cruising altitude) from the atmospheric properties function used in the aircraft sizing tool, except dynamic viscosity. The length values are approximated from the scaled drawings in [] (MAC has been assumed to be the characteristic length for the wings and tails). All of these are listed in Table 4.4.

Parameter	Value
Density of air (ρ)	0.3653 kg/m^3
Velocity (V)	250.89 m/s
Cruise Mach (M)	0.85
Dynamic viscosity (μ)	1.5 Ns/m^2
Fuselage length (l)	61.37 m
Wing MAC length (c)	12.65 m
Horizontal tail MAC length (MAC)	6.33 m
Vertical tail length	9.49 m
Nacelle characteristic length	6.14 m

Table 4.4: Inputs for calculating skin friction coefficient and form factor

Using (18), Reynold's number for every component of the aircraft is to be calculated individually and the values obtained are given in Table 4.5.

Component of aircraft	Reynold's number
Fuselage	3.63×10^8
Wing	6.57×10^7
Horizontal tail	3.286×10^7
Vertical tail	4.929×10^7
Nacelle	3.286×10^7

Table 4.5: Reynold's numbers of air around aircraft components

All values in table 4.2 represent a turbulent flow boundary layer and the corresponding skin friction coefficient formula is given by (19).

$$C_f = \frac{0.455}{[(\log Re)^{2.58} + (1 + 0.144 \times M \times n^2)^{0.65}]} \quad (19)$$

- Re - Reynold's number
- M - cruise Mach number
- n - load factor

Here, the load factor is 1, as the flight is assumed to be steady.

Using (19), skin friction coefficient for every component of the aircraft can now be calculated individually and the values obtained are given in Table 4.6.

Component of aircraft	Skin friction coefficient
Fuselage	0.0016582
Wing	0.0020955
Horizontal tail	0.0023288
Vertical tail	0.0021844
Nacelle	0.0023188

Table 4.6: Skin friction coefficients for aircraft components

Estimating F for every component

The form factor is an empirical value that accounts for the pressure (from) drag. Similar to skin friction coefficient, form factor must be calculated individually for each component of Le Plane 381.

1. Fuselage:

$$F = 1 + \frac{2.2}{\lambda^{1.5}} - \frac{0.9}{\lambda^3} \quad (20)$$

- $\lambda = \frac{l}{(\frac{4}{\pi} \times A)^{0.5}}$
- l - length of fuselage
- A - cross sectional area of fuselage
- $Q = 1.0$ - interference factor

2. Wing:

$$F = (F^* - 1) \times \cos^2(\lambda_{0.5c}) + 1 \quad (21)$$

$$F^* = 1 + 3.3 \left(\frac{t}{c} \right) - 0.008 \left(\frac{t}{c} \right)^2 + 27.0 \left(\frac{t}{c} \right)^3 \quad (22)$$

- $\lambda_{0.5c}$ - sweep angle at 50% chord
- $\frac{t}{c}$ - tip to chord ratio
- $Q = 1.1$

3. Tail surfaces:

$$F^* = 1 + 3.52 \left(\frac{t}{c} \right) \quad (23)$$

- $Q = 1.2$

The calculated values of form factor are listed in Table 4.7.

Component of aircraft	Form factor
Fuselage	1.0513
Wing	1.2283
Horizontal tail	1.3300
Vertical tail	1.2750

Table 4.7: Form factor values for aircraft components

Estimating Q (interference factor)

The interference factor accounts for the interference drag which is generated by the mixing of airflow streams between different airframe components.

The values of Q for the fuselage, wing and tail surfaces were taken from Jenkinson [15]. Jenkinson gives a value for the product of the form factor and the interference factor of a nacelle, $FQ = 1.25$.

Component of aircraft	Interference factor
Fuselage	1.0
Wing	1.0
Horizontal tail	1.2
Vertical tail	1.2

Table 4.8: Interference factor values for aircraft components

Estimating ratio of wetted area to reference area

Wetted area is the area which is in contact with the external airflow and reference area is the planform area of the wing.

All wetted areas were calculated using scaled drawings of the reference aircraft used for our missions (Boeing 767-400ER).

To calculate the fuselage wetted area, the 2D drawing found was first circumscribed inside a rectangle. Assumptions to account for the shape of the front and back of the fuselage were made and a ratio of this area to the area of the rectangle was calculated. The fuselage was then perfectly circumscribed inside a cylinder, and its total surface area was calculated. Assuming proper equivalence between the 2D shapes and their 3D counterparts, the ratio was multiplied by the total surface area of the cylinder and the value obtained was taken as our fuselages wetted area.

The wing and horizontal tail wetted area was calculated using the formula $A = b \times c$ where b is the wingspan/tail span and c is the MAC.

The vertical tail was first idealised as a trapezium and the area of this was taken as our vertical tails wetted area.

The nacelle was idealised as a complex 3d shape containing a cylinder and frustum. The nacelle waited area was then calculated using the appropriate curved surface area formulae for a cylinder and a frustum.

The calculated values are listed in Table 4.9.

Component of aircraft	$\frac{S_{wet}}{S_{ref}}$
Fuselage	3.6999
Wing	1.0184
Horizontal tail	0.0499
Vertical tail	0.1810
Nacelle	0.4246

Table 4.9: Form factor values for aircraft components

All of these values were put into the formula for $C_{D,0}$ (17) and then summed up to give the final $C_{D,0}$ value for the aircraft. All of these are listed in Table 4.10:

Component	C_f	F	Q	$\frac{S_{wet}}{S_{ref}}$	$C_{D,0}$	Amount	Final $C_{D,0}$
Fuselage	0.001658232	1.0513	1	3.6999	0.006450033	1	0.006450033
Wing	0.0020955	1.2283	1	1.0184	0.002621262	2	0.00524252
Horizontal tail	0.002318836	1.33	1.24	0.04988	0.000190752	2	0.0003815
Vertical tail	0.002184357	1.275	1.24	0.181	0.000625078	1	0.000625078
Nacelle	0.002318836	1.25		0.42457	0.001230635	2	0.00246126
						$C_{D,0}$	0.015160391

Table 4.10: Form factor values for aircraft components

5 Landing gear

Landing gear is a key aspect of aircraft design, as it is used to absorb the energy due to the impact upon landing. This minimises the loads transmitted to the airframe, meaning no injuries to the passengers and crew and no damage to the aircraft. The obvious advantage is that landing gear allows the plane to takeoff, land and taxi safely but it also contributes to the longitudinal stability of the aircraft, which prevents tipping and tilting.

The position of the main landing gear is given by (24).

$$x_{mg} = \frac{x_{cg} - kx_{ng}}{1 - k} \quad (24)$$

- x_{mg} - position of main landing gear
- x_{cg} - centre of gravity
- k - load weight fraction of front landing gear

- x_{ng} - position of front landing gear

The value k is a constant that represents the ratio between the load experienced by the front landing gear and the total weight of the aircraft. The nose landing gear for commercial aircraft are generally designed to take 10% of the aircraft's total weight, therefore a value of $k = 0.1$ was used. The front landing gear position was chosen to be 5.84 m aft of the top of the nose. This value was attained by considering factors such as similarities to the reference aircraft and space in the fuselage for landing gear retraction. Inputting these values of k and x_{ng} and the previously calculated centre of gravity value outputs the position of the main landing gear to be 33.75 m aft of the tip of the nose.

The landing gear track is the distance between the centrelines of the main landing gears. For these, a 2 by 3, 6-wheel landing gear configuration was chosen due to increased stability stemming from a wider landing gear track. This occurs because the weight of the aircraft can be distributed over a larger area which is what reduces the chance of tipping or swaying. The landing gear wheelbase is the distance between the centrelines of the front and main landing gear, which, as can be calculated from Figure 1 is around 27.9 m. This affects stability in the same way that the track does, with a longer wheelbase being ideal for balance and stability.

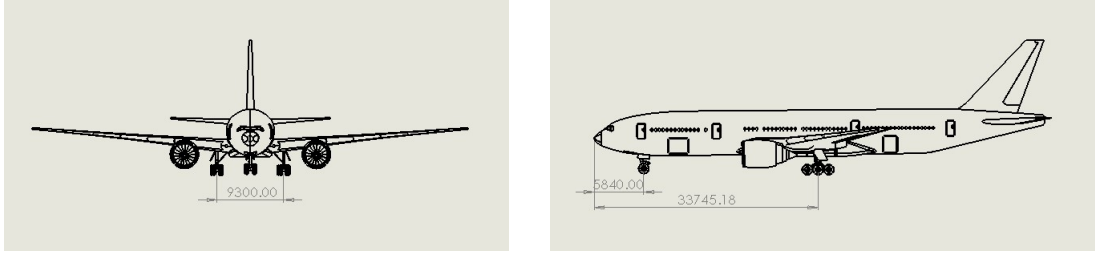


Figure 16: Landing gear track (left) and landing gear wheelbase (right)

5.1 Front Landing Gear

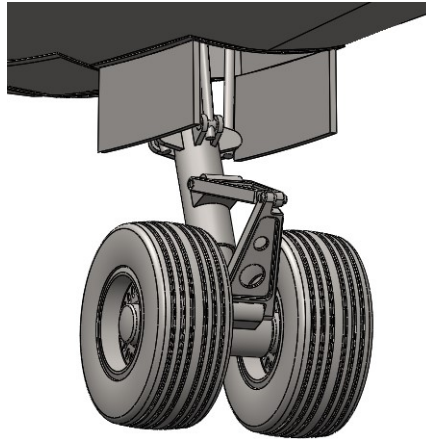


Figure 17: Front landing gear 3D view

Figure 17 shows the opening in the fuselage into which the landing gear will retract.

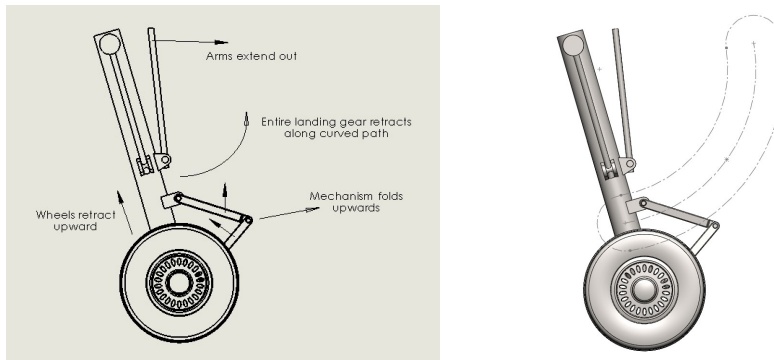


Figure 18: Front landing gear mechanism explanation (left) and front landing gear retraction path (right)

The retraction process starts with the main mechanism folding together upwards, causing the wheel to retract upwards. Then, the arm extends out and up, converting linear motion into rotary motion. This in turn causes the landing gear to follow the retraction trajectory shown in Figure 18 (right) backwards into the fuselage. In this figure, it shows this trajectory starting above the centre of the wheel. This is because the main mechanism will have already folded up at this point, meaning the wheel will be higher up the shaft before it begins to follow the arc into the body of the aircraft.

5.2 Main Landing Gear

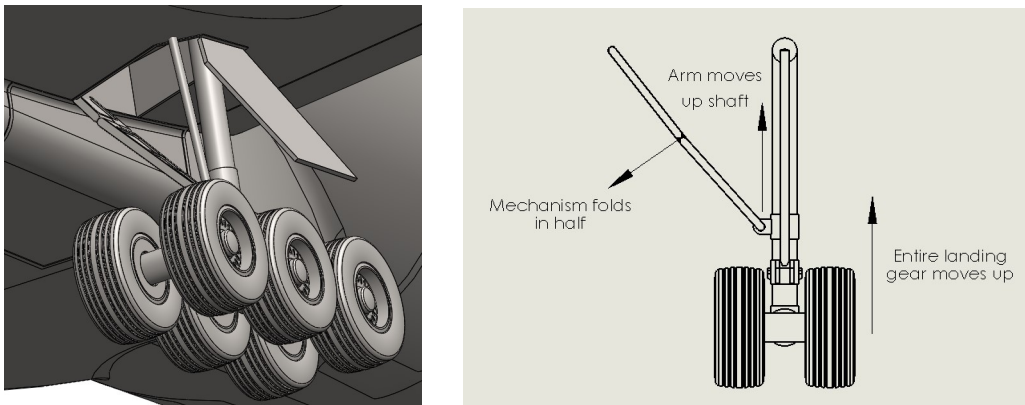


Figure 19: Front landing gear mechanism explanation (left) and front landing gear retraction path (right)

The main landing gear's retraction method needed to be different to that of the front landing gear to accommodate for the larger number of wheels and therefore an increase in weight. The main mechanism is larger and folds together outwards instead of inwards like for the front landing gear. Following this, the arm moves up the shaft, taking the landing gear up with it into the fuselage of the aircraft. Therefore, its retraction path is simply straight up.

6 Market Analysis

Air travel trends following COVID-19

Air travels trends are important to discuss because they change every year due to various factors including GDP growth, fuel prices, technological advances, environmental conditions, natural events and government regulations.

Trends in air travel always exist and are constantly changing, however one whose effects have lasted for a prolonged period of time was when the COVID-19 pandemic hit. The global economy went into crisis, leading air travel to reduce significantly. COVID-19 was an unprecedented disruption, leading to the number of passengers on commercial flights to reduce by 60.2% compared to 2019 (before the pandemic). The demand for international and domestic travel dropped by 75% and 49% respectively.

A loss of approximately 372 billion USD to this sector was observed in 2020 alone. Many airlines experienced major losses and some declared bankruptcy. By October 2020, 43 airlines were bankrupt (this number has since increased). Later in October 2020, ACI Europe declared that 193 airports out of 740 in Europe were at risk of bankruptcy.

Air Italy ceased their operations, Turkish Airlines collapsed in February 2020, challenges were faced by many including German Airways, Thai Airways, City Jet, Air Mauritius, etc. It was a tough time for everyone and the recovery is still being made to this day. Figure 20 demonstrates the decline in commercial airliners.

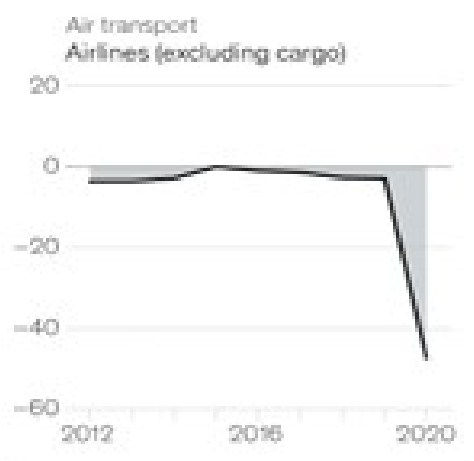


Figure 20: Decline in commercial airlines

All the parts of the world had undergone losses in 2020 due to the pandemic. North America, who earned significant profits from 2012-2019 lost 63 billion USD in 2020, as seen in Figure 21.

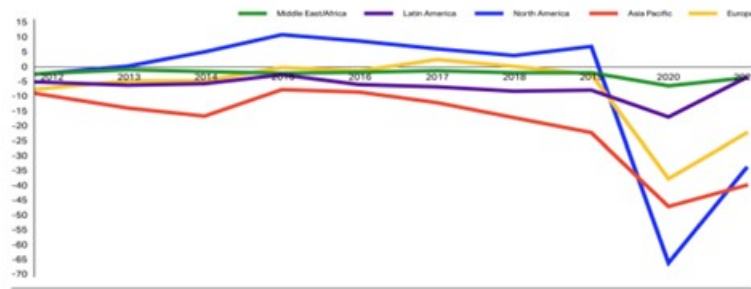


Figure 21: Economic losses of continents

Aircraft orders went down by 54% in 2020 but came back up by 154% in 2021. The pandemic also resulted in numbers of leased narrow-bodied aircraft going from 42% in 2012 to 51% in 2021 and leased widebodies from 27% to 35%. This period caused airlines to go for options like leasing and sale. Airlines restructured leases by adopting hourly arrangements and this was done by the ones that were on the brink of bankruptcy.

The original equipment sector had earned an ROIC of about 16% from 2012 to 2019 and during the pandemic, ROIC dropped to -26% in 2020 and -2% in 2021, as shown in Figure 22.

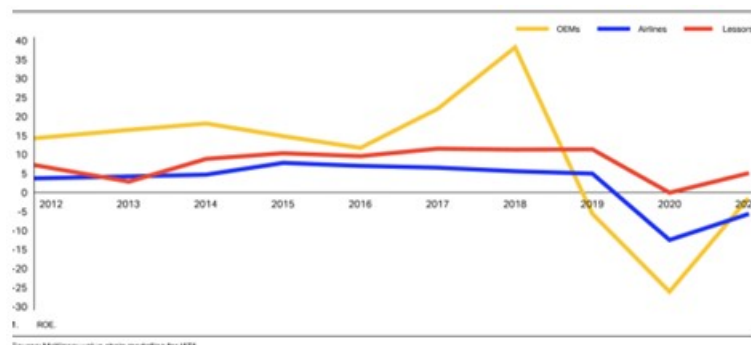


Figure 22: ROIC trends

Before the pandemic, sectors such as catering, ground hauling, maintenance, repair and operations were doing well and the ground hauling sector saw rises in ROIC by 16% between 2012-2019, as shown in Figure 23. When air traffic decreased drastically in the pandemic this ROIC dropped to -7%. For catering this rise was 20% from 2012-2019 and then dropped to -21% in 2020 and -16% in 2021. There was noticeable improvement in the MRO sector after the pandemic, with a ROIC of 3% in 2021 from -5% in 2020.

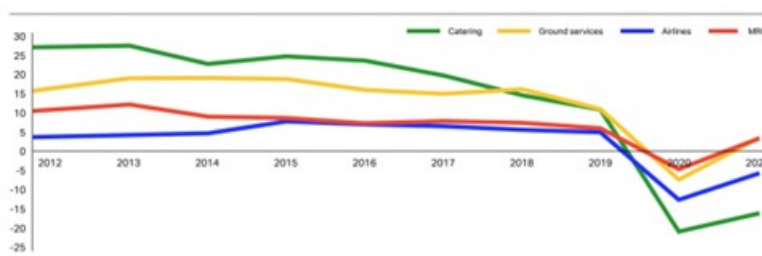


Figure 23: ROIC in different sectors from 2012-2021

One of the few positive things during the pandemic was increase in the cargo airliners.

Freight forwarders also saw major improvements and profitability, shown in Figure 24.

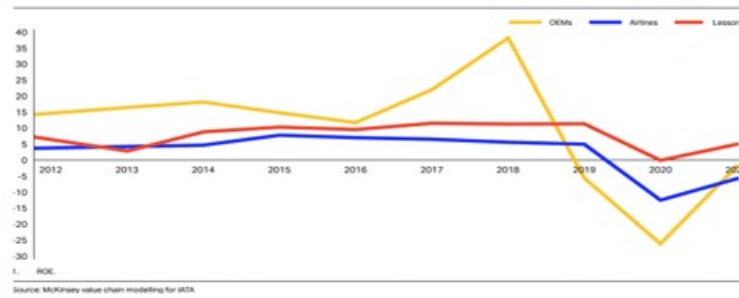


Figure 24: ROIC trends, 2012-2021

Flight cancellations also became more frequent and underwent increases during January 2020 (Figure 25).



Figure 25: Flight cancellation trends

Over the years, air traffic has evolved, risen and fallen down because of various events that have impacted it. This is portrayed in Figure 26.

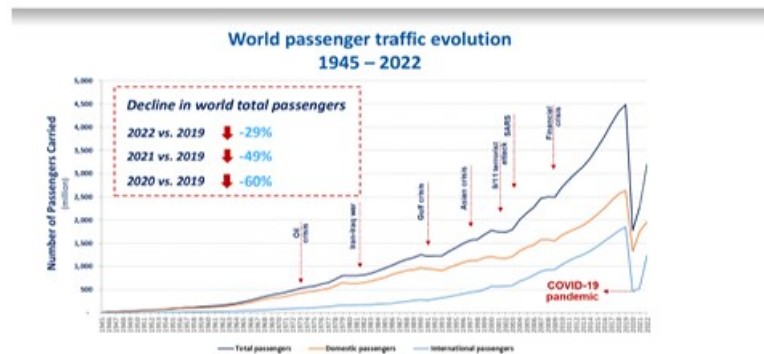


Figure 26: Air traffic trends, 1945-2022 [16]

North America overtook China and ended up having a 30% contribution to domestic aircraft traffic. The trends across the world were as shown in Figure 27.

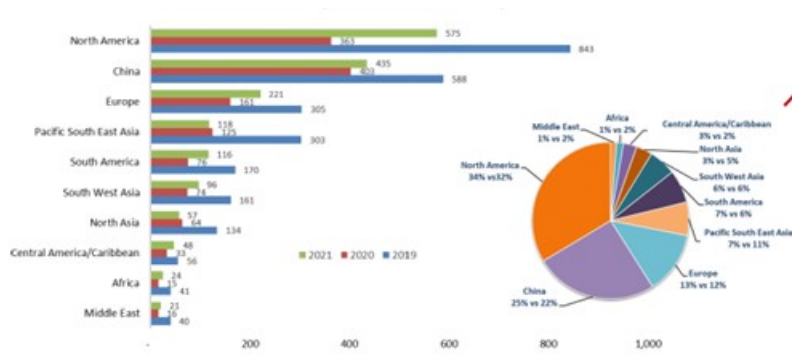


Figure 27: World air traffic trends [17]

How COVID-19 effects were managed

One of the factors which led to the recovery of the aviation market after the pandemic was the support that was provided by the government to various airlines. Governments distributed financial aid packages to keep airlines from going bankrupt and ensuring air connectivity.

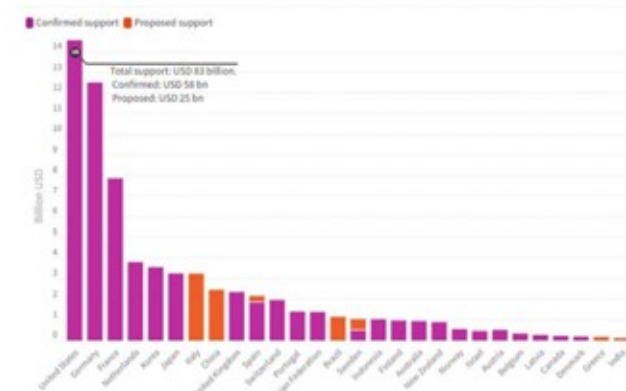


Figure 28: Government support to different countries after COVID-19 [18]

The implementation of safety measures and health protocols has been a top priority and following cleaning procedures, mask-wearing and enhanced ventilation systems have ensured safe air travel.

Digital transformation has also been one of the factors that has helped recover the aviation market. Airlines and airports have adopted new technologies to continue operations and improve customer service. Increased use of mobile apps for booking, check-ins and boarding has reduced physical contact and has provided more convenience.

The aviation industry also increased its cargo operations to diversify revenue streams. This has helped generate more income and has helped relieve pandemic challenges.

COVID-19 has also made airlines consider the issues of sustainability and hence the effort to use more fuel-efficient aircraft has been brought up. Governments and the aviation industry have focused on reducing their carbon footprint, to make the industry more eco-friendly.

Routes have been optimized and airlines have focused on routes that have shown

improvements in air traffic and finances during the pandemic.

Collaborations and partnerships among various airlines, airports and other bodies have worked in alignment to establish health and safety measures.

The world is still navigating the uncertainties after the pandemic, meanwhile the aviation industry remains committed to provide and ensure safe and sustainable air travel. All these past challenges will lead the industry to have a more innovative future.

Current and future air travel

Many of the macroeconomic factors like global inflation, slowdown of GDP, business confidence levels and political conflicts pose challenges and uncertainties in future forecasts.

In 2024, the aviation market size is expected to reach 336.96 billion USD and will continue to grow at a CAGR of 2.87% and is expected to reach 396.15 billion USD by 2029. The key players in the aviation market have been Boeing, Lockheed Martin and Airbus.

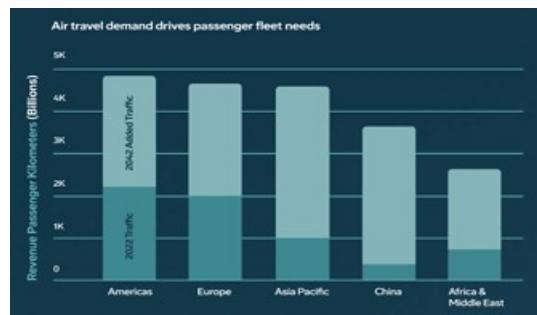


Figure 29: Air travel demands [19]

Air travel is meant to continuously increase due to a fast growing rate of tourism. Global passenger air travel has become immensely diverse and balanced over the last few decades and this will continue over next 20 years. Mature aviation sectors like North America and Europe are expected to undergo a slower growth rate in comparison to emerging aviation sectors in Asia, as these regions are in high growth phases of infrastructure and airline development.

Passenger traffic is expected to reach 3.9 billion passengers by the end of 2025 in the Asia-Pacific region. In Europe and North America, it is expected to reach 2.7 billion and 2.3 billion respectively, as seen in Figure 30. The growth has been rapid in Asian regions because of the ongoing developments in those regions.



Figure 30: Passenger traffic forecast

As shown in Figure 31, in 2052, global passenger traffic is expected to reach 25 billion which is nearly 2.5 times the 2024 forecast.

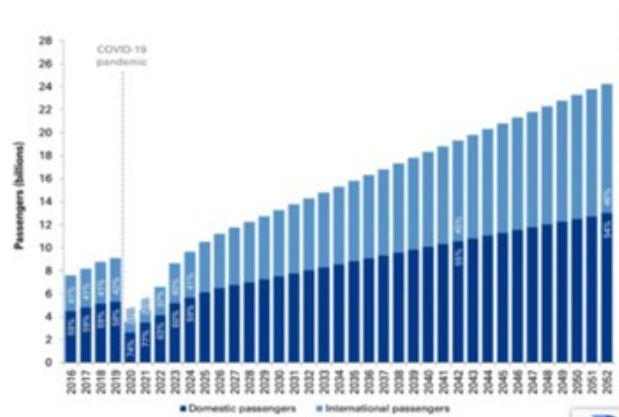


Figure 31: Global passenger traffic

The aviation industry keeps growing and with the ongoing development, an increase in these trends is expected.

Air travel trends: Manchester International Airport (EGCC)

Manchester International Airport is the third busiest Airport in the United Kingdom and 19th busiest in the whole of Europe, with 28.1 million passengers being served. It has flights flying to 199 different destinations. It went through a major expansion of terminal 2 in 2017 and will serve 35 million passengers by 2024. The capacity is actually 50 million passengers, however there are rules and regulations which state that only 61 movements per hour are allowed.

The airport has three terminals and all of them are linked to each other via skylink and travelators. Terminal 1 is used by airlines with operations flying to Europe and other destinations. By 2025, the airlines operating from terminal 1 will shift to Terminal 2. Terminal 2 is capable of extensive expansion and will be able to handle 25 million passengers a year. It's the most developed terminal with new piers and bigger security halls. The transformation programme announced in January 2023 that the third phase will include the refurbishment of the existing terminal and will introduce a new security hall. It includes

the construction of pier 2 and it is aimed to be completed by 2025. Terminal 3 is the new domestic terminal and serves domestic flights.

The work on the VIP terminal of the airport began in 2019 and it opened in October 2019. This terminal has been closed since COVID-19 and is meant to open in Autumn 2024.

In the last decade, Monarch Airlines, Thomas Cook Airlines and Flybe all ceased operations and this impacted local employment and flight operations, leaving thousands of passengers stranded whilst many were abroad. Immense efforts were made to save Flybe, which was operating half of the UK domestic flights but nothing worked and it finally ceased operations in early 2020. Data which was published by the Civil Aviation Authority (CAA) noticed a drop of passengers from 29,374,282 in 2019 to 6,787,127 in 2020. The reduction led to terminal 2 and 3 completely getting shut down during that period. This decline is clearly shown in Figure 32.

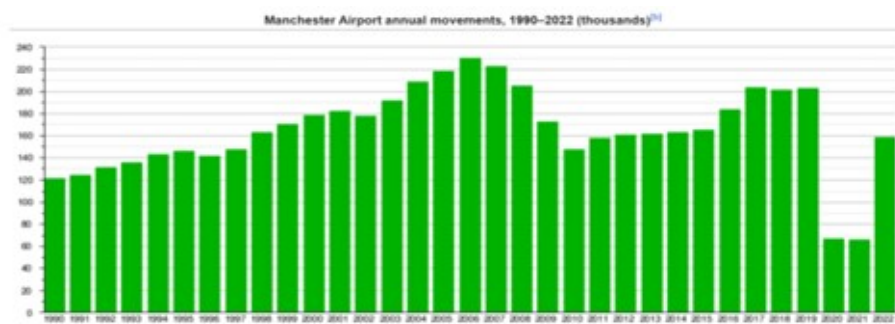


Figure 32: Manchester Airport annual movements [20]

The future of Manchester International Airport [21] includes continuous expansion until 2030. Demolition of older buildings has begun to provide a new taxiway and Eastwards extension of terminal 2, which is meant to provide 15 more covered standards. The development plans are being done to meet the ongoing and increasing demand for flying. According to the CAA, the number of passengers that the Airport served was 28,077,659 passengers and this was higher than the previous year which was 23,364,471. The recovery from COVID-19 is still being made, as it served 29,397,357 passengers in 2020. Airlines like Etihad Airways has one of its six maintenance bases in Manchester. Until September 2019, it was also the engineering base of Thomas Cook Airlines.

Air travel trends: Los Angeles Airport (KLAX)

This Airport serves Los Angeles and the nearby metropolitan cities of California, United States. It is situated in Westchester, a neighbourhood city of LA and is 18 miles southwest of LA. It is closest to Westside and South Bay and is operated by Los Angeles World Airports (LAWA) which is a branch of the Los Angeles City government. It has four parallel runways and covers 35,000 acres of land.

In 2022, the airport operated 65,924,298 passengers and became the worlds sixth busiest airport. It was also in top ten busiest airports in 2023. It is a major international gateway

in the United States, providing connections to passengers travelling to East and Southeast Asia, Mexico, Central America and Australasia. This airport holds the record of being worlds busiest origin and destination airport, as a lot of passengers begin or end their journey here.

In 2019, LAWA reported that the airport had 88% of its passengers beginning or ending their journey and remaining 12% were on connecting flights. It serves as an operating base for more passenger airlines than any other airport in United states.

The Los Angeles airport has 150 gates and 9 passenger terminals. By the end of 2025, the airport will have an automated people mover, which is meant to link other terminals and provide connections to parking facilities. Additionally, the airport got cargo facilities that cover an area of 2 million square feet.

The airport had plans to become more modern at a cost of 30 billion USD and this will increase the total number of gates from 142 to 186.

The air traffic at this airport has kept it at the top of the list of busiest airports and it served 87 million passengers in 2018. It had a passenger volume of 88,068,013 in 2019, which dropped down to 28,779,527 in 2020 when the pandemic hit. In 2023, it recorded a passenger volume of 75,050,851 showcasing the improvement from 2020 but the recovery in the numbers is still being made.

The busiest international route is to London Heathrow, UK with 1,546,657 passengers in September 2023. The other busiest ones include Guadalajara, Mexico; Vancouver, Canada; Tokyo, Japan and Sydney, Australia. The busiest domestic routes include New York JFK, San Francisco and Las Vegas.

The largest irlines at LAX is Delta Airlines, serving 14,831,038 passengers annually, according to a Report in 2023. The other major airlines have been American Airlines, United Airlines, Southwest Airlines, Alaska Airlines, Air Canada and Hawaiian Airlines.

Air travel trends: Tahiti International Airport (NTAA)

Tahiti international Airport (also called Faaa International Airport) is situated 3.1 miles southwest of Papeete. It is the only international Airport in French Polynesia. It serves both international and domestic flights. Its the hub for Air Tahiti and Air Tahiti Nui. It has daily flights to other islands in French Polynesia and one international service to the Cook Islands.

Intercontinental flights to Canada, Chile, Metropolitan France, United States, Japan and New Zealand are available. The airport is on an island and has limited level terrain. It has a single runway and can accommodate aircraft like Boeing 747 and Airbus A380. It recorded an annual number of passengers of 1,708,098 in 2023. The airlines operating here are Air France, Air Moana, Air New Zealand and Delta Airlines. The airport plays a prominent role in the tourism industry of French Polynesia and tourism contributes to 12% of GDP for

French Polynesia. Tourists have contributed to the countrys economic growth and in 2019 it had 300,000 tourists but in 2020, this number dropped to 89,400 due to the restrictions of the pandemic.

Air travel trends: Orlando International Airport (KMCO) [22]

Orlando International Airport is the primary international airport in Florida which is 6 miles southeast of Orlando. The airport is the hub for silver airways and the operating base for Avelo Airlines, JetBlue and Southwest Airlines. It is one of the largest commercial airports in the United States. Its the major gateway for the Florida region, having 850 flights daily on 44 airlines. It serves 135 domestic and international destinations. Untill 2007, it was also the hub of Delta Airlines.

The airfield was originally constructed to serve the US Army Air Force in the early 1960s when jet airline flights were introduced to Orlando and it became a joint civil-military facility. The 1971 opening of Magic Kingdom at Walt Disney wWorld led to a significant increase in its air travel as it became a major tourist destination. There are very few enclaves that the military still uses in the airport.

The airport came to be a US Customs service for the foreign trade zone in 1978. It handled 7 million passengers in 1971 and this number increased to 47 million in 2018. It covers an area of 47 square kilometres and is the fifth largest airport in the United States.

It has three terminals with a total of 93 gates. It serves various airlines including British Airways, Avianca, Air Canada, Air Transat and American Airlines. The busiest domestic route of the Airport in 2023 was to Atlanta, Georgia with 1,452,000 passengers. The busiest international route was to Toronto in 2022 with 77,656 passengers.

The airport recorded 57,735,726 passengers and this was 15.1% higher than the previous year. In 2020, the annual number of passengers dropped down to 57% from 2019, recording only 21,617,803 passengers.

6.1 Substantiation of Future Value from Today

Le Plane 381 flies 1560 passengers per week from EGCC to NTAA and 312 passengers per week from KLAX to KMCO. This aircraft is designed, such that it will carry 200 passengers per flight, with 8 flights per week from EGCC to NTAA and 2 flights per week from KLAX to KMCO. The number of flights is low, which saves fuel consumption per passenger, leading to more profits for airlines.

Le Plane 381 is suitable for the medium long-haul flights and long-haul flights. During the pandemic, air travel trends were negatively impacted, which it resulted in an immense

number of losses financially. These regions (EGCC, NTAA, KLAX and KMCO) demonstrate optimum growth in the future for the air travel demands and suggest an increasing number of passengers, leading to greater profits. The decision to carry 200 passengers per flight is made to support the maximum MTOW limits of the aircraft. Not only does this allow the MTOW to be in a suitable range but it also benefits the airlines to meet good RPKs and maintain the value of factors such as TSFCs and fuel consumptions per passenger.

This analysis demonstrates how these regions are going to grow their passenger demand and how Le Plane 381 will produce good profits and optimum results, meeting passenger demands and ensuring their safety as well.

In addition to passenger demand, carrying cargo could be considered on these routes, as with the economic picking back up after COVID-19, the demand for cargo services on these routes is forecast to increase. Market research shows that the demand for air travel between Europe and the South Pacific region, as well as the United states will increase.

7 Additional Sections from Tables 2 & 3

7.1 Aircraft Compatibility Information

Airport compatibility refers to the ability of an airport to accommodate various types of aircraft. This includes various considerations like the size of the aircraft that that airport can accommodate, infrastructure availability at airports to support the aircraft and any restrictions that might apply. The market analysis provides insights on how demands of air travel went down for a few years during the pandemic and how, with a great deal of technological advancements and increasing number of passengers, these compatibility factors are currently changing across the globe.

Many airports worldwide have started operations to build new runways to accommodate larger aircraft and increased frequency of passengers. This part of the report solely focuses on the airports of the chosen trajectories and provides a detailed compatibility analysis of them.

Table 7.1 mentions the 4 airports that the trajectories include. Three of these (EGCC, KLAX, NTAA) form the primary route and two (KLAX, KMCO) form the secondary route.

ICAO CODE	EGCC	NTAA	KLAX	KMCO
No. of Runways	2	1	4	4
Runway length (metres)	3200	3420	2,721-3,939	2,743 - 3,659
Runway width (metres)	45	45	60	61
Elevation (metres)	78	2	39	29

Table 7.1: Technical information for this mission’s airports

Manchester Airport (EGCC) is the start point of the primary route and it encapsulates 4 runways and is located at an elevation of 78 m. It has two runways, both of which can be simultaneously operated for take-off and landing operations. It has three terminals which are linked to each other via travelators. From 2017 to 2024, the airport went through major expansion projects which led terminal 2 to be doubled in size and this enabled the terminal to accommodate 35 million passengers.

This airport can accommodate various types of aircraft, ranging from A380s to private jets, such as the Embraer E-jet. Since Le Plane 381’s wingspan is smaller than an A380’s this airport will easily accommodate this aircraft.

In case of emergency circumstances, there are two nearby airports that could serve as diversions and assist in the smooth landing of the aircraft. These include Liverpool John Lennon Airport (EGGP) and Leeds Bradford Airport (EGNM). Two other airports which are closer in proximity to EGCC are Manchester Barton Aerodrome (EGCB), Blackpool Airport (EGNH) and Hawarden airport (EGNR). Unfortunately, their runway length is insufficient and would not accommodate Le Plane 381. EGPP and EGNM are 30 miles southwest and 45 miles northeast of Manchester, so it would be easy for the plane to divert to these two airports if needed. However, depending on the nature of the emergency, fuel remaining onboard and other aircraft capabilities, a diversion to London Heathrow Airport (EGLL), which is 160 miles southeast of Manchester Airport, could also be made.

The primary route of the aircraft is a long-haul flight stretching to a flight time of nearly 18 hours. Due to this, a decision to make a stop at KLAX was made for refuelling and passenger disembark/embark purposes. The airport has four runways and 9 passenger terminals with 150 gates. The airport is undergoing terminal 4 and 6 construction, which will provide accommodation for the increasing number of passengers in the coming years.

The airport handles a diverse mix of aircraft sizes and models, including A380, A350 XWB, 787 Dreamliner, 777, 747-400 etc. The Los Angeles airport has undergone significant infrastructure upgrade and expansion. It is one of the busiest and most important airports in the world and accommodated 75 million passengers in 2023 with a total of 542,749 aircraft movements in that year alone. In case of emergency landings, the nearby airports which might serve as diversion destinations are Ontario International Airport (KONT), Long Beach Airport (KLGB), Burbank Hollywood Airport (KBUR) or John Wayne Airport (KSNA). All of these airports can accommodate Le Plane 381 logistically and physically.

The destination of the primary route is Tahiti International Airport (NTAA). It is operated by Seti Aero ports and has a single runway capable of accommodating a Boeing 747. In 2023, the airport accommodated 1,708,098 passengers and with the tourism demands rising again in the region, the air travel demands in this region are also expected to go up. It is the primary airport in French Polynesia and serves the island of Tahiti. There are several smaller airports in French Polynesia, but none in proximity that can accommodate wide-body aircraft such as Le Plane 381. However, a few airports in French Polynesia can handle Boeing 767-400 ER for emergency landings or stops if and when NTAA isn't available for landing. Their primary purpose is to serve domestic flights within French Polynesia and have limited infrastructure compared to NTAA but are the only options in case of emergencies. These are Moorea Airport (MOZ), Raiatea Airport (RFP) and Bora Bora Airport (BOB). These airports are not international hubs but they have runways and facilities that can assist in case of emergency landings.

The only viable location for a diversion en route between Los Angeles and Tahiti is the Honolulu/Daniel K. Inouye International airport (KHNL), which at any point in the route is at least 850 nautical miles away. This is why Le Plane 381 needs to accommodate a lot more fuel in the place of passengers. It has 4 major runways, 60 gates and 3 terminals. All of its runways have lengths of between 2,120 and 3,753 m, meaning there will be enough LDA. It saw around 23 million passengers in 2023, which means it boasts plenty of facilities for airplanes and passengers alike.

The origin and destination of the secondary route are both based in the United States. KMCO has four runways and terminals. It is one of the busiest airports in Florida and the USA. In 2021, it saw 19,618,838 flights, which made it the busiest airport in the US and 7th busiest in the world. Orlando is a world-renowned tourism destination and every year it attracts millions of tourists to its theme parks and resorts. It is a major hub for business and leisure travel as well.

The airport is building a new terminal which is south of the terminal A/B complex. When the construction is finished, the south terminal complex is going to have 120 new gates. Phase 1 of the South Terminal Complex includes new aircraft taxiways and aprons. Construction of Terminal D will begin when the airport reaches 70 million passengers annually. When the construction is done, the airport will have capacity of 100 million passengers annually. It can accommodate a wide range of commercial aircraft, from narrow body aircraft like the A318 to wide-body aircraft like the Boeing 777. This means that the airport will easily accommodate Le Plane 381. Nearby airports include Orlando Sanford International Airport (KSFB), Daytona Beach International Airport (KDAB), Melbourne Orlando International Airport (KMLB) and Tampa International Airport (KTPA).

7.2 Flight Deck Sight Angles

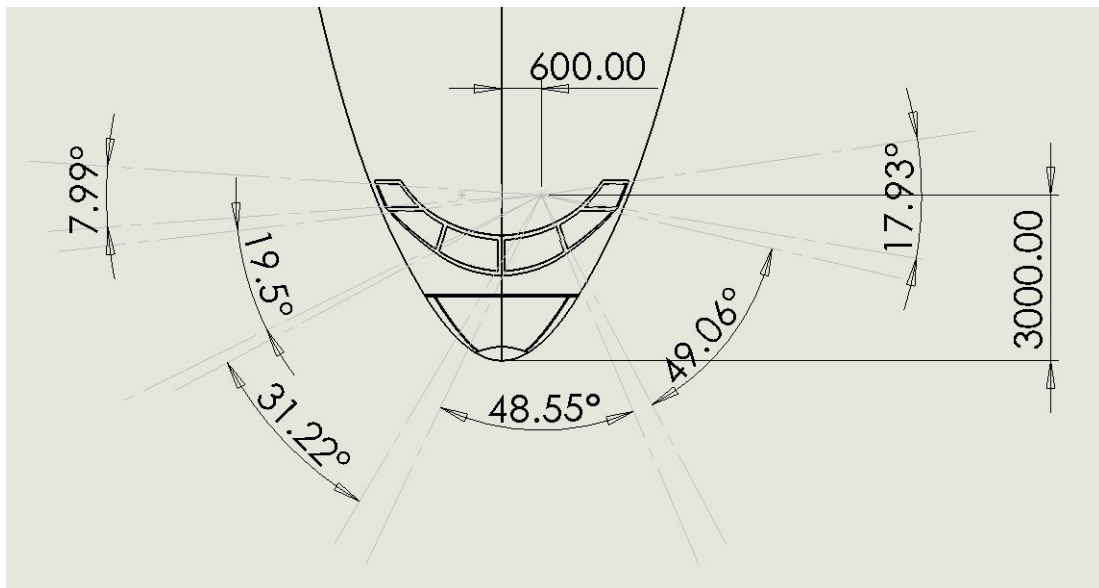


Figure 33: Flight Deck Sight Angles SOLIDWORKS Top View

Flight deck sight angles are crucial for pilots to be able to see other aircraft while flying. It is important to design the windows of the cockpit in a way that ensures that there are no blind spots so that the pilots can safely perform manoeuvres within the operating limits of the aircraft.

It was assumed that there are two pilots who are sitting 1.2 m away from each other and both are 3 m away from the tip of the nose cone. They are indicated in Figure 1 by two construction points. This aircraft has 6 windows and therefore 6 sight angles, however, the view from furthest window from each pilot will be blocked by the other pilot. For the case of Figure 1, this means the 7.99° angle should not be considered. Since the angles for only one pilot are shown, it might seem worrying that there are some blind spots where the fields of view do not overlap, however, once the second pilot is considered, this issue will not exist, as they can make up for what the first pilot cannot see.

In the event that at any point during the mission, only one pilot remains, there are eye position gauges for pilots to adjust their seats to achieve the best view.

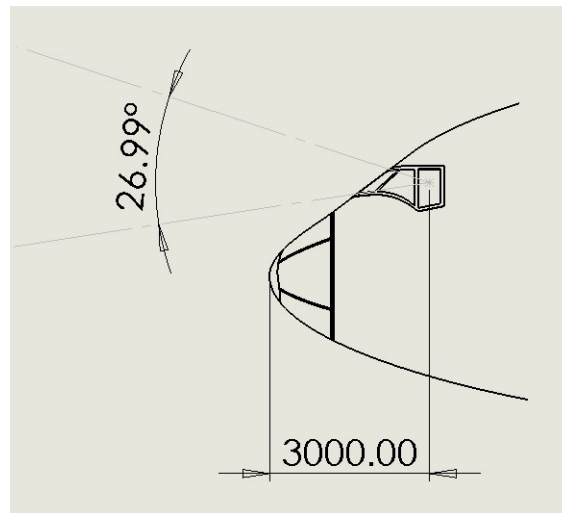


Figure 34: Flight Deck Sight Angles SOLIDWORKS Side View

7.3 Ground Servicing and Timing Plans

Aircraft turnaround is an important procedure and it is defined as the process of preparing the aircraft for its next flight. This includes disembarking and embarking of passengers as the plane lands and gets ready for its next take-off. The turnaround time of the aircraft varies from 45-120 minutes and this depends on the type of aircraft and the type of flight, ie whether it is a short-haul flight or long-haul flight.

The aircraft turnaround time has to be established in limits of ground-servicing, which is around 45-90 minutes [23]. The efficiency of ground servicing depends on various factors, including availability of staff, scheduling, equipment, performance and meteorological conditions. The turnaround time of an aircraft is directly related to ground servicing.

The turnaround preparation begins with the process of data sharing. Messages are received and then actions are taken in accordance to them. These messages use common international language to give details about their destination. These messages include information about the number of passengers and the estimated time of arrival.

After touchdown, the aircraft is directed to a specific stand which offers facilities to accommodate it and help it perform the necessary ground servicing operations, so that it can efficiently be prepared for its next scheduled departure. As the aircraft comes to a standstill, the turnaround process is initiated. This involves the disembarking of passengers which is done through airbridges, buses or walking from the aircraft to the terminal.

Cleaning of the aircraft is also done by the contracted companies. A good amount of cleaning staff is required, varying in number and requirement according to the aircraft and flight duration. Delay in the cleaning procedure can lead to further delays in turnaround time. Catering operations are also carried out and this involves the removal of rubbish from near-empty galleys and replacing them with full ones for the next flight.

Loading and unloading of bags is a vital step in the whole process and is done by two units, which are the operation department and the ramp department. The most important step is the fuelling one, which is done with the assistance of fuel trucks or a hydrant fuelling system and these are located on each of the parking stands. Maintenance checks are also made which ensure the aircrafts worthiness for the next scheduled flight.

Pre-flight checks are also a major step in ground servicing and this involves cockpit inspections, engine inspections and flight control decks. After each of these is carried out efficiently, the boarding process can start. For it to start, the necessary communication between ramp, traffic and cabin crew occurs, which confirms that all the turnaround processes have been carried out. Once all of these pre-flight operations are finished, the aircraft is pushed back and positioned for taxiing to runway. It receives take-off clearance from the ATC (air traffic control) and can depart.

Ground servicing has to be carried out effectively and with attention as a lag in it, can hinder further processes. The devotion of staff members is crucial and their effective performance improves the turnaround. Facilities being provided at the airports also affect these ground handling activities. For example, Singapore Changi Airport is ranked among the top airports globally and has efficient ground services because it provides automated check-ins, fast security clearances and efficient baggage handling.

With ongoing technological advancements, the turnaround process will become faster due to incorporation of advanced sensors, AR and VR reality, mobile applications, remote monitoring and control, robotics and drones. Implementing automation into aircraft ground handling systems has also been considered, as this will provide the opportunity to reduce fuel consumption, reducing workload for the crew. For this to be possible, redesign of the ground facilities is necessary but with the pace of technological developments, all of this automation seem possible for the near future. By 2050, turnaround time for commercial aircraft is expected to become faster and with increasing demand of air travellers, this will be beneficial [24].

Le Plane 381's reference aircraft is a Boeing 767-400ER and it will follow the same turnaround process, which will take an average of 90 minutes to carry out. However, Boeing 767-400ER holds 245 passengers, and the plane designed here incorporates 200 passengers, so the turnaround time could be less.

7.4 Vn diagram

The V-n diagram is a flight envelope that indicates the load factor and velocity limits of the aircraft under which it wont acquire any structural damage. The velocity at different load factors was calculated from a rearranged version of (25).

$$n_{max} = \frac{1}{2}\rho V^2 \frac{C_{L,max}}{\frac{W}{S}} \quad (25)$$

Commercial aircraft experience typical load factors between the range of -1 and 2.5, so these were the load factors taken here.

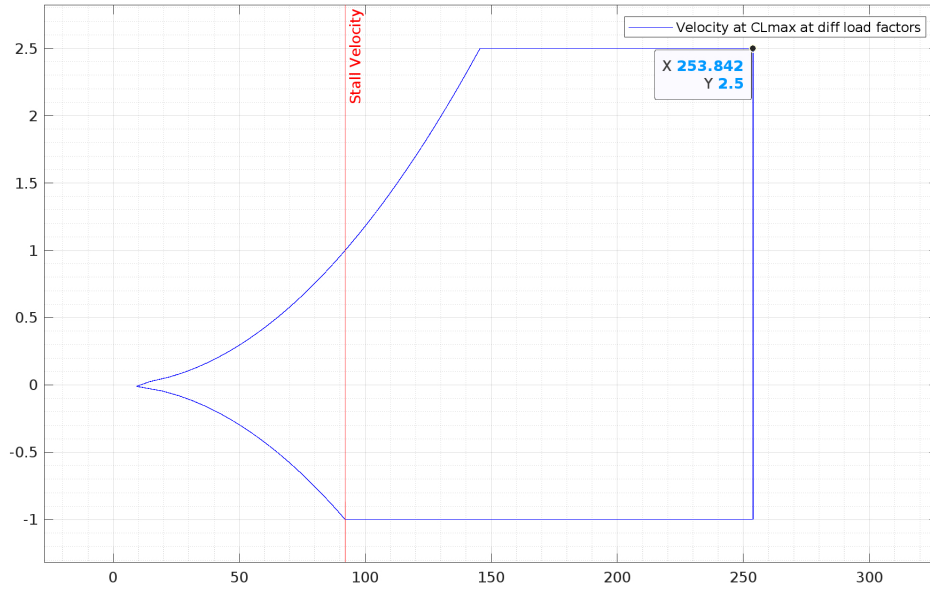


Figure 35: Vn diagram for Le Plane 381

The curved part of the plot until it meets the two horizontal lines illustrates the velocity limits at different load factors and the maximum lift coefficient used here is the maximum lift coefficient at maximum lift to drag ratio. The vertical line which bounds the right part of the graph, marks the maximum dive velocity and was assumed to be Mach 0.86 after referring to many similar aircraft and flights dive velocities and maximum operating mach number. The x line marks the stall velocity which falls right on the limit of our V-n graph.

As mentioned before, the weight estimation tables, the payload range chart and constraints analysis chart were all produced with the assumption that the cruise velocity is Mach 0.85. So, it is very clear that Le Plane 381 will have plenty of room for manoeuvres and will be able to fly at any speed or load factor that a typical commercial flight might need.

8 Appropriate neutral point and cg located on drawings

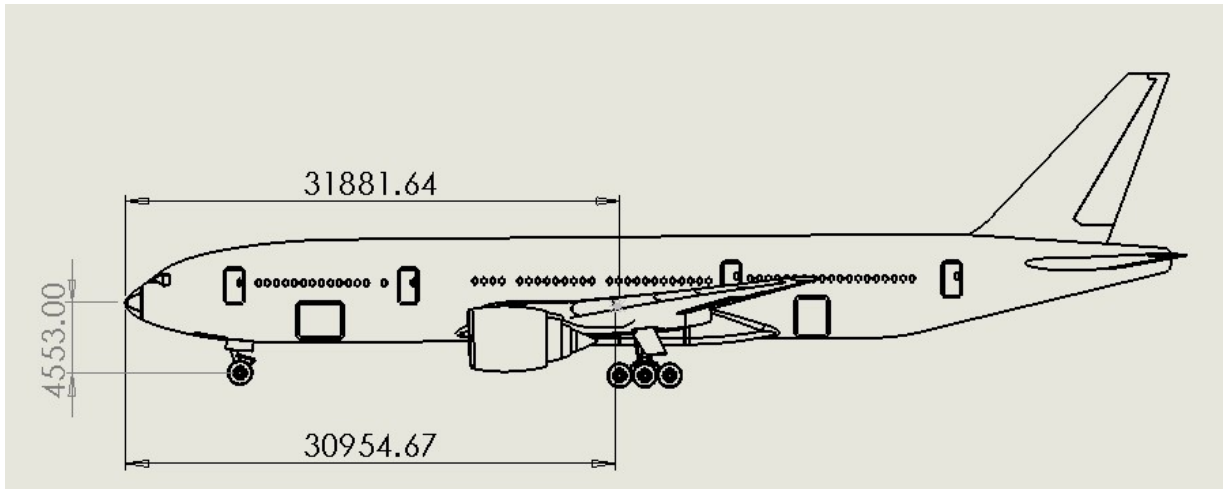


Figure 36: cg and neutral point location

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